STATISTICS · UNIT 1: EXPLORING DATA

Lunch Line Data Detectives

Use displays, summary statistics, normal models, and regression to fix the school lunch line.

Honors

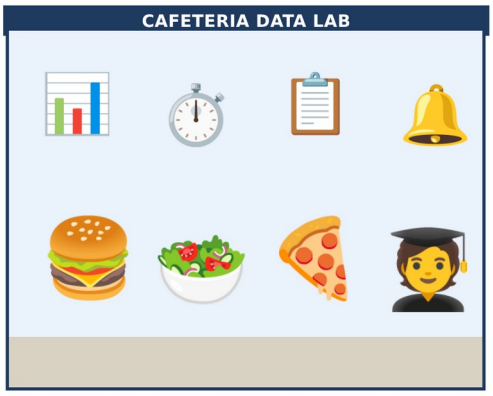
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Students say the lunch line is too slow, and the principal wants evidence before changing anything. Your team has been given data from the cafeteria: wait times at the Grill line and the Salad Bar, lunch choices by grade, and how wait time depends on the number of people in line.

You will describe, compare, and model the data, then recommend one change backed by statistics.



Grill Line

Median wait 8 min

One wait of 25 min!

Salad Bar

Median wait 6 min

IQR 3 min

Learning Goals

- I can choose and justify summary statistics for skewed and symmetric data.
- I can use normal calculations, percentiles, and conditional distributions.
- I can build, assess, and critique a least-squares model using residuals and influence.

Project Timeline

When	What you will do
Day 1	Distributions
Day 2	Normal Model and Tables
Day 3	Regression and Influence
Final	Data Report to the Principal + reflection

Part 1: Distributions

Day 1 · Use the Grill and Salad wait times on the Data Cards.

Cafeteria Data Cards

Grill waits (min)	Salad waits (min)	Semester model
4, 5, 6, 6, 7, 7, 8, 8,	3, 4, 4, 5, 5, 6, 6, 6,	Daily average Grill wait
8, 9, 10, 11, 12, 14, 25	7, 7, 7, 8, 8, 9, 10	Normal, mean 9, SD 2 min

Choice by grade	Line length x	Wait time y
Gr 9: Grill 30, Salad 10, Pizza 40	2, 4, 5, 6, 8, 10, 12, 15	3, 4, 6, 6, 8, 9, 11, 13
Gr 12: Grill 24, Salad 26, Pizza 30	people in line	minutes

1. Give the five-number summaries, means, and SDs of both lines, and identify outliers.

2. The cafeteria adds 2 minutes to every Grill wait to account for paying. What happens to the median, IQR, mean, and SD?

3. The times are converted to seconds. Find the new median and SD of the Salad Bar.

4. Write a full comparison and argue which measure of center and spread the principal should use for each line.

Part 2: Normal Model and Tables

Day 2 · Daily average Grill wait is approximately N(9, 2) minutes.

5. Find the proportion of days with averages between 8 and 11 minutes.

6. The principal wants to flag the slowest 10% of days. Find the cutoff.

7. Make segmented bar graphs of lunch choice by grade and find P(senior | salad).

8. If grade and choice were not associated, how many seniors would be expected to choose salad? Compare with the data.

Part 3: Regression and Influence

Day 3 · Use the line-length (x) and wait-time (y) data.

9. Use $\bar{x} = 7.75$, $\bar{y} = 7.5$, $s_x \approx 4.367$, $s_y \approx 3.423$, and $r \approx 0.994$ to find the LSRL by formula.

10. Interpret the slope, the y-intercept, r^2, and the residual at $x = 10$ ($y = 9$).

11. A new point (30 people, 8 min) is added. Is it an outlier, a high-leverage point, or influential? Predict how it changes the slope and r.

12. Describe what a residual plot would need to look like for the linear model to be appropriate.

Final Product: Data Report to the Principal

Your final product must include:

- Full distribution comparison with justified statistics
- Effects of shifting and rescaling
- Normal calculations and the slow-day cutoff
- Conditional distributions and expected counts
- Regression by formula with an influence analysis and a recommendation

Reflection

R1. Why does adding a constant change the center but not the spread?

R2. How could one unusual point mislead the principal about the line-length model?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Displays, summaries, and models describe the lunch data correctly and support the recommendation. Goes beyond the minimum with a creative, realistic choice.	Displays, summaries, and models describe the lunch data correctly and support the recommendation.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes displays, statistics in context, and one clear recommendation for the principal. Polished, organized, and visually engaging.	Report includes displays, statistics in context, and one clear recommendation for the principal. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.